There are three morphological patterns in ER.

1. Granular or Rough endoplasmic reticulum
2. Smooth endoplasmic Reticulum
3. Lamellar and Vesicular endoplasmic reticulum

The rough endoplasmic reticulum contains ribosome attached to the cytoplasmic side of the membrane and it forms a lace like system. The smooth Endoplasmic reticulum lacks the attached ribosome and it forms tubular structures.

The major functions of Endoplasmic reticulum are:

- They play a vital role in the formation of the skeletal framework
- They provide the increased surface area for cellular reactions
- They help in the formation of nuclear membrane during cell division
- They play a vital role in the synthesis of proteins, lipids, glycogen and other steroids like cholesterol, progesterone, testosterone, etc.
- They are responsible for the secretion, synthesis, modification and transportation of proteins and other carbohydrates to another organelle, which includes lysosomes, Golgi apparatus, plasma membrane, etc.

1.4.6.Golgi Apparatus

Camillo Golgi (1898) had made the first report on the densely stained reticular structures near the nucleus. Hence these were later named Golgi bodies, attributed to him. They consist of many flat, disc-shaped sacs or cisternae of 0.5µm to 1.0µm diameter. These are stacked parallel to each other. Varied numbers of cisternae are present in a Golgi complex. The Golgi cisternae are concentrically arranged near the nucleus with distinct convex *cis* or the forming face and concave *trans* or the maturing face.

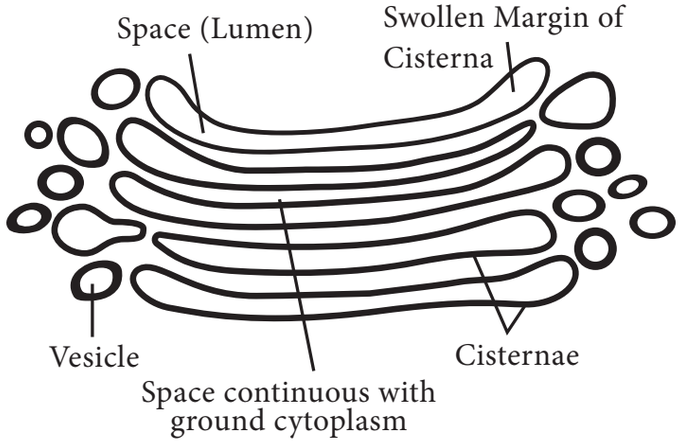

Figure 1.9.A schematic diagram of Golgi apparatus

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